

Sandeep Kumar Amgothu

Data Engineering | Machine Learning & AI Safety | Cloud & MLOps | Analytics & Power Platform

✉ sandeepkumaramgothu3@gmail.com | 📞 +1 (203) 997-4125 | [in LinkedIn](#) | [GitHub](#) | [Portfolio](#)
📍 Richmond, Virginia

August 15, 2026

Hiring Manager
Company Name
Team/Department
City, State

Re: [Job Title] — [Company Name]

Dear Hiring Manager,

Your need for an engineer who can ship production ML systems that are scalable, observable, and safe maps directly to how I deliver. I am a Machine Learning and Data Engineer with 5+ years spanning LLM security and automated red teaming, cloud-native ETL/ELT, computer vision, and MLOps—bridging research rigor with enterprise deployability. Most recently I have driven nonprofit analytics platforms at Data Pioneer while continuing to harden LLM evaluation pipelines that cut attack success rates by 20–50 percentage points across 3,000+ adversarial interactions.

At Texas A&M University–Corpus Christi, I led an automation-first, taxonomy-driven red-teaming platform that orchestrates GPT-4o-mini as a red-team agent against quantized Llama-3.2-1B-Instruct, with ensemble safety graders (LlamaGuard, DeBERTa reward, Detoxify, RoBERTa) producing composite L0–L3 harm scores and replayable logging. In parallel, I engineered a UAV audio detection MLOps pipeline processing 50,000+ acoustic samples—cutting STFT/Mel feature extraction from 15 hours to under 5 hours with Spark/Python—and trained CNN/RNN/CRNN/VGG19 models to 97.2% F1-score and 90%+ accuracy at sub-100 ms inference, published at ICUAS 2025.

At Tata Consultancy Services (Jul 2021–Jul 2023), I architected AWS ETL pipelines (S3, Redshift, Glue, Lambda) processing 15M+ legal and financial records per month for a Fortune 500 client (Xerox), improving latency by 40% and sustaining 99.9% data availability for 500+ enterprise users, while automating 10K+ document ingestions monthly and lifting downstream ML accuracy by 15% through feature and data-quality frameworks.

I am drawn to [Company Name] because [1–2 specific sentences: mission, product, tech stack, or recent initiative from the JD/company site]. Roles where hands-on engineering meets thoughtful design of safety, reliability, and user impact are where I contribute most—whether the priority is securing LLMs, building robust ML platforms, or delivering high-quality data products under real operational constraints.

I would welcome a 20-minute technical demo or call to walk through a recent red-teaming pipeline, MLOps design, or cloud ETL/Power BI solution aligned to your stack. Thank you for your time and consideration.

Sincerely,

Sandeep Kumar Amgothu

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[linkedin.com/in/sandeepkumaramgothu](https://www.linkedin.com/in/sandeepkumaramgothu)

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Microsoft Certified: Power BI Data Analyst Associate | Azure Administrator Associate | Power Platform Developer Associate